Unit 6 Exercises

4SC000 Optimal Control and Reinforcement Learning

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A runnable Python companion to these exercises – a general discrete-time Kalman filter (prediction + correction), reused from `code/unit4/`, plus the mass-spring-damper example of Problem 6.3 – is available in `code/unit6/` (see its `README.md`).

Core exercises

Problem 6.1 The goal of this exercise is to repeat Problem 5.1 but considering state disturbances and noisy measurements. To this effect consider a first order discrete-time linear system

$$x_{k+1} = \alpha x_k + u_k + w_k, \quad k \in \mathbb{N}_0,$$

where $\{w_k\}$ are Gaussian, independent and identically distributed, with variance $\mathbb{E}[w_k^2] = \sigma_w^2$, and consider the cost

$$\mathbb{E}\Big[\sum_{k=0}^{q} x_k^2 + \gamma u_k^2\Big].$$

(a) Considering $\alpha = 1$, $\gamma = 1$, $q = 2$, $\sigma_w = 1$, determine the policy for u_0, u_1, u_2 that minimizes the cost when: (i) the value of x_k is known at every stage k. (ii) x_k is measured at every stage by a noisy sensor $y_k = x_k + n_k$, where $\{n_k\}$ are Gaussian iid with variance $\mathbb{E}[n_k^2] = \sigma_n^2$, $\sigma_n = \frac{1}{3}$, and $x_0 \sim \mathcal{N}(0, 10)$.

(b) Considering $\alpha = 1$, $\gamma = 1$, and the averaged cost $\lim_{q\to\infty} \frac{1}{q+1}\mathbb{E}[\sum_{k=0}^{q} x_k^2 + \gamma u_k^2]$, determine the policy for $\{u_k\}$ that minimizes the cost when: (i) x_k is known at every stage. (ii) x_k is measured as in (a)(ii), with the same distribution for x_0.

Solution

(a)(i) By certainty equivalence, the policy is the same as in Problem 5.1(a): $u_2 = \mu_2(x_2) = 0$, $u_1 = \mu_1(x_1) = -\frac{1}{2}x_1$, $u_0 = \mu_0(x_0) = -\frac{3}{5}x_0$.
(a)(ii) Let $I_0 = \{y_0\}$, $I_1 = \{u_0, y_0, y_1\}$, $I_2 = \{u_0, y_0, y_1, u_1, y_2\}$. By the separation principle, the optimal policy replaces the state with its Kalman-filter estimate:

$$u_2 = \mu_2(I_2) = 0 \qquad u_1 = \mu_1(I_1) = -\tfrac{1}{2}\hat{x}_{1|1} \qquad u_0 = \mu_0(I_0) = -\tfrac{3}{5}\hat{x}_{0|0}$$

Since $u_2 = 0$, we only need L_0 and L_1. With $\hat{x}_{0|-1} = \bar{x}_0 = 0$, $\Phi_{0|-1} = 10$, $A = B = C = 1$, $W = 1$, $V = \frac{1}{9}$, iterating $\Phi_{k+1|k} = A\Phi_{k|k}A^\top + W$, $\Phi_{k|k} = \Phi_{k|k-1} - \Phi_{k|k-1}C^\top(C\Phi_{k|k-1}C^\top + V)^{-1}C\Phi_{k|k-1}$, $L_k = \Phi_{k|k-1}C^\top(C\Phi_{k|k-1}C^\top + V)^{-1}$:

$$\Phi_{0|0} = 10 - \frac{10^2}{10 + \frac{1}{9}} = \frac{10}{91}, \quad L_0 = \frac{10}{10 + \frac{1}{9}} = \frac{90}{91} \qquad \Phi_{1|0} = \frac{10}{91} + 1 = \frac{101}{91}, \quad L_1 = \frac{101/91}{101/91 + 1/9} = 0.909$$

so $\hat{x}_{0|0} = \frac{90}{91}y_0$, $\hat{x}_{1|0} = \frac{90}{91}y_0 + u_0$, $\hat{x}_{1|1} = \hat{x}_{1|0} + 0.909(y_1 - \hat{x}_{1|0})$.

(b)(i) For the average-cost problem the policy coincides with the deterministic infinite-horizon policy from Problem 5.1(b): $u_k = -0.618 x_k$.
(b)(ii) By the separation principle, $u_k = -0.618\hat{x}_{k|k}$, where $\hat{x}_{k|k}$ is now the *stationary* Kalman filter estimate (appropriate for an average-cost criterion, where the initial transient is irrelevant): $\hat{x}_{k+1|k} = \hat{x}_{k|k} + u_k$, $\hat{x}_{k|k} = \hat{x}_{k|k-1} + L(y_k - \hat{x}_{k|k-1})$, $L = \frac{\Phi}{\Phi+1/9}$, where Φ is the positive solution of $\Phi^2 = \Phi + \frac{1}{9}$, i.e. $\Phi = 1.1009$ (discarding the negative root -0.1009), giving the stationary gain $L = 0.9083$.

Problem 6.2 Consider the system

$$x_{k+1} = \frac{5}{6} x_k + u_k + w_k$$

with open-loop policy $u_k = 1$ for every $k \in \mathbb{N}_0$, and w_k zero-mean iid Gaussian with $\mathbb{E}[w_k^2] = 1$. We wish to estimate x_k based on the measurements

$$y_\ell = 2x_\ell + \tfrac{1}{2} v_\ell, \quad \ell \in \{0, 1, \dots, k\}$$

where v_ℓ are zero-mean iid Gaussian with $\mathbb{E}[v_\ell^2] = 2$, and $x_0 \sim \mathcal{N}(\frac{1}{2}, 1)$.

(i) Show that the Kalman filter estimate $\hat{x}_{2|2} = \mathbb{E}[x_2 | y_0, y_1, y_2, u_0{=}1, u_1{=}1]$ can be written as $\hat{x}_{2|2} = a_0 y_0 + a_1 y_1 + a_2 y_2 + c$, and give a_0, a_1, a_2, c.
(ii) Show that the stationary estimate $\hat{x}_{k|k} = \mathbb{E}[x_k | y_0, \dots, y_k, u_0{=}1, \dots, u_{k-1}{=}1]$ can be written as $\hat{x}_{k|k} = \sum_{\ell=0}^{k} h_{k-\ell} y_\ell + d_k$, and give an expression for h_k and d_k.

i Solution

Not reproduced here – taken from the TU/e 2015/2016 sample exam 1, whose full worked solution is available in the exam archive rather than in SolutionsPS2.tex. The approach follows the standard Kalman filter recursion (prediction with $A = \frac{5}{6}$, $B = 1$, $W = 1$; correction with $C = 2$, $V = \frac{1}{2}$) applied twice by hand for (i), then generalized to the stationary (converged) gain for (ii), expressing each $\hat{x}_{k|k}$ as a linear combination of past outputs – structurally the same computation as Problem 6.5, with a nontrivial A and C.

Additional exercises

Problem 6.3 Consider a mass-spring-damper system for which the mass displacement $z(t)$ is described by

$$m\ddot{z}(t) = -c_f \dot{z}(t) - k_s z(t) + f_d(t), \quad t \in \mathbb{R}_{\geq 0}$$

where $m = 1$, $k_s = 5$, $c_f = 1$, and $f_d(t)$ models force disturbances.

(a) Find a state-space representation considering $x(t) = [z(t) \;\; \dot{z}(t)]^\top$.
(b) Find a discrete-time system describing $x(k\tau)$ for $\tau = 0.1$, approximating $f_d(t) \approx f_d(k\tau)$ where $f_d(k\tau)$ are zero-mean iid Gaussian with $\mathbb{E}[f_d(k\tau)^2] = 0.1$.
(c) Suppose the output $y(k\tau) = Cx(k\tau) + n_k$ is available, n_k zero-mean iid Gaussian with $\mathbb{E}[n_k^2] = \sigma_n^2$. (i) Provide the Kalman filter estimate of $x(k\tau)$ for $k \in \{0, 1, 2\}$, assuming the initial state covariance is $2I$. (ii) Provide the stationary Kalman filter and the asymptotic error covariance matrix.

Solution

A structurally identical problem (same physical setup, mass-spring system with position measurements) is worked out in full in the course slides – see Lecture 6, Appendix C ("Example: Kalman filter", lectures/lecture06-kalman-lqg.qmd), which discretizes an analogous mass-spring system, sets up W and V, and iterates the Kalman filter equations by hand for $k = 0, 1, 2$; follow the same steps here with $m = 1$, $k_s = 5$, $c_f = 1$ in place of the lecture's parameters.

Problem 6.4 Consider the discrete-time linear system

$$x_{k+1} = Ax_k + Bu_k + \omega_k, \quad k \in \mathbb{N}_0 \qquad A = \begin{bmatrix} 1 & \frac{1}{5} \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(a) Suppose $\omega_k = 0$ for every k. Compute the optimal u_0, u_1, u_2 minimizing $J = \sum_{k=0}^{h-1} x_k^\top Q x_k + u_k^\top R u_k$ for $h = 3$, $Q = I$, $R = 0.1$.

(b) Suppose ω_k are zero-mean iid Gaussian with $\mathbb{E}[\omega_k \omega_k^\top] = \frac{1}{2} I$ and the full state is available. Compute the optimal policy $u_k = \mu_k(x_k)$, $k \in \{0, 1, 2\}$, minimizing $\mathbb{E}[J]$.

(c) Suppose ω_k are as in (b) but only $y_k = Cx_k + n_k$ is available, $C = [1 \;\; 0]$, n_k zero-mean iid Gaussian with $\mathbb{E}[n_k^2] = \frac{1}{4}$. (i) Compute the optimal policy $u_k = \mu_k(I_k)$, $k \in \{0, 1, 2\}$, minimizing $\mathbb{E}[J]$. (ii) Compute the optimal policy minimizing the average cost $\lim_{h\to\infty} \frac{1}{h}\mathbb{E}[J]$.

(d) Suppose now the actuators can only be updated at even times $k = 2\ell$ (multi-rate control), $u_k = 0$ at odd k. (i) Compute the optimal policy $u_k = \mu_k(I_k)$, $k \in \{0, 2\}$, minimizing the cost of (c)(i). (ii) Compute the optimal policy for $k = 2\ell$ minimizing the average cost of (c)(ii).

Solution

(a) This is a standard (deterministic) finite-horizon LQR problem; solving the Riccati recursion with $P_3 = 0$ gives $P_2 = I$, $P_1 = \begin{bmatrix} 2 & 0.2 \\ 0.2 & 1.1309 \end{bmatrix}$, $K_2 = 0$, $K_1 = [0 \;\; -\frac{1}{1.1}]$, $K_0 = [-0.1625 \;\; -0.9513]$. Propagating $x_0 = [1 \;\; 1]^\top$ through the dynamics with $u_k = K_k x_k$ gives $u_0 = -1.1137$, $x_1 = [1.2 \;\; -0.1137]^\top$, $u_1 = 0.1034$, $u_2 = 0$.

(b) By certainty equivalence, the optimal policy is identical to (a): the disturbances change the *cost* but not the optimal gains K_0, K_1, K_2 (since x_0 is deterministic, u_0 is known a priori, but u_1 depends on the realization of w_0 through x_1).

(c)(i) By the separation principle, $u_k = K_k \hat{x}_{k|k}$ with the same gains as (a)-(b), and $\hat{x}_{k|k}$ the Kalman filter estimate ($\hat{x}_{0|-1} = \bar{x}_0 = [1 \;\; 1]^\top$, $\Phi_{0|-1} = 0$ since x_0 is known exactly, $W = \frac{1}{2} I$, $V = \frac{1}{4}$). Iterating gives $\Phi_{0|0} = 0$, $L_0 = 0$ (so $\hat{x}_{0|0} = x_0$ regardless of y_0), $\Phi_{1|0} = \frac{1}{2} I$, $L_1 = \frac{2}{3}[1 \;\; 0]^\top$. Since $K_2 = 0$ and $K_1 L_1 = 0$ (the second component of K_1 multiplies a zero entry of L_1), it turns out u_0, u_1, u_2 all end up *deterministic* – numerically identical to the noise-free path in (a): $u_0 = -1.1137$, $u_1 = 0.1034$, $u_2 = 0$. This is a special feature of this particular A, B, C, not a general fact about output feedback.

(c)(ii) By the separation principle for the average-cost problem, $u_k = K \hat{x}_{k|k}$ with K the infinite-horizon LQR gain and $\hat{x}_{k|k}$ from the *stationary* Kalman filter. Solving the two algebraic Riccati equations (control and estimation) numerically gives $P = \begin{bmatrix} 6.6069 & 1.2044 \\ 1.2044 & 1.3506 \end{bmatrix}$, $\Phi = \begin{bmatrix} 0.8781 & 0.7510 \\ 0.7510 & 3.4230 \end{bmatrix}$, $L = \begin{bmatrix} 0.7784 \\ 0.6658 \end{bmatrix}$, $K = [-0.8303 \;\; -1.0971]$.

(d)(i)-(ii) Grouping two time-steps into one ($\bar{x}_\ell := x_{2\ell}$, $\bar{u}_\ell := u_{2\ell}$) turns the multi-rate problem into a standard single-rate problem with $\bar{A} = A^2$, $\bar{B} = AB$, $\bar{Q} = Q + A^\top Q A$, $\bar{R} = R + B^\top Q B$, $\bar{S} = A^\top Q B$, and lumped disturbance $\bar{w}_\ell = A w_{2\ell} + w_{2\ell+1}$ (still Gaussian iid, since sums of Gaussians are Gaussian). For (d)(i) (finite horizon, $h = 3$) this gives the single-decision gain $\underline{K}_0 = [-0.0935 \;\; -0.9720]$ and, since $\hat{x}_{0|0} = x_0$ exactly, $u_0 = -1.0654$ independent of y_0. For (d)(ii) (average cost) the same reasoning with the algebraic Riccati/Kalman equations for $(\bar{A}, \bar{B}, \bar{Q}, \bar{R}, \bar{S})$ gives $\bar{P} = \begin{bmatrix} 6.6274 & 1.2045 \\ 1.2045 & 1.3536 \end{bmatrix}$, the same Φ, L

as (c)(ii), and gain $\bar{K} = [-0.7905 \;\; -1.1269]$.

Problem 6.5 Consider the problem of estimating the value of a constant x_0 from noisy measurements

$$y_k = x_0 + n_k, \quad k \in \mathbb{N}_0,$$

where n_k are zero-mean iid Gaussian with variance $\mathbb{E}[n_k^2] = \sigma_n^2$. The initial state x_0 is also Gaussian with mean $\bar{x}_0$ and variance $\phi_0 = \mathbb{E}[(x_0 - \bar{x}_0)^2]$.

(i) Compute the Kalman filter estimate $\hat{x}_k$ of x_0 (the one minimizing the error variance $\mathbb{E}[(x_k - \hat{x}_k)^2]$) at times $k \in \{0, 1, 2\}$, based on $I_0 = \{y_0\}$, $I_1 = \{y_0, y_1\}$, $I_2 = \{y_0, y_1, y_2\}$, and the associated error variances. [Suggestion: note $y_k = x_k + n_k$ for $x_{k+1} = x_k$.]

(ii) Compute the asymptotic error variance $\lim_{k\to\infty} \mathbb{E}[(x_k - \hat{x}_k)^2 | I_k]$ and explain the value obtained.

i Solution

(i) Since $x_{k+1} = x_k$, $y_k = x_k + n_k$, the disturbance covariance is $W = 0$. The standard Kalman filter equations reduce to $\hat{x}_{k+1|k} = \hat{x}_{k|k}$, $\hat{x}_{k|k} = \hat{x}_{k|k-1} + L_k(y_k - \hat{x}_{k|k-1})$, where $L_k = \frac{\phi_{k|k-1}}{\phi_{k|k-1}+\sigma_n^2}$ and $\phi_{k+1|k} = \phi_{k|k} = \frac{\sigma_n^2 \phi_{k|k-1}}{\phi_{k|k-1}+\sigma_n^2}$. Writing $\hat{x}_k := \hat{x}_{k|k}$, $\phi_k := \phi_{k|k-1}$ (note $\phi_k = \phi_{k|k-1} = \phi_{k|k} = \mathbb{E}[(x_k - \hat{x}_k)^2 | I_k]$):

$$\hat{x}_0 = \bar{x}_0 + \frac{\phi_0}{\phi_0 + \sigma_n^2}(y_0 - \bar{x}_0) = \bar{x}_0 \frac{\sigma_n^2}{\phi_0 + \sigma_n^2} + y_0 \frac{1}{\phi_0 + \sigma_n^2}$$

a weighted average of the prior mean and the first measurement (weights set by the ratio of ϕ_0 to σ_n^2). Then $\hat{x}_1 = \hat{x}_0 + \frac{\phi_1}{\phi_1+\sigma_n^2}(y_1 - \hat{x}_0)$ and $\hat{x}_2 = \hat{x}_1 + \frac{\phi_2}{\phi_2+\sigma_n^2}(y_2 - \hat{x}_1)$, where $\phi_1 = \frac{\sigma_n^2 \phi_0}{\phi_0+\sigma_n^2}$, $\phi_2 = \frac{\sigma_n^2 \phi_1}{\phi_1+\sigma_n^2}$; $\hat{x}_1$ is a linear combination of y_0, y_1 (and $\hat{x}_2$ of y_0, y_1, y_2).

(ii) $\lim_{k\to\infty} \mathbb{E}[(x_k - \hat{x}_k)^2 | I_k] = \lim_{k\to\infty} \phi_k$. Since $\phi_k > 0$ and $\phi_{k+1} < \phi_k$ for every k, the limit is zero: as k grows we accumulate more and more independent measurements of the *same* constant, so our estimate's uncertainty is averaged away to nothing.

Problem 6.6 Suppose we wish to estimate the state, at a given time $k = h$, of a linear system $x_{k+1} = Ax_k + w_k$, $k \in \{0, \dots, h-1\}$, with A invertible so that $x_k = A^{-1}x_{k+1} - A^{-1}\bar{w}_{k+1}$, $\bar{w}_{k+1} := w_k$. Output measurements $y_k = Cx_k + v_k$ are available for $k \in \{0, \dots, h\}$, and an initial estimate $\bar{x}_0$ of x_0 is available (but not the true x_0).

One way to estimate x_h (denoted $\bar{x}_h$) is to (i) solve the optimal control problem

$$J_h(x_h) = \min_{\bar{w}_1, \dots, \bar{w}_h} (x_0 - \bar{x}_0)^\top \bar{\Theta}_0^{-1}(x_0 - \bar{x}_0) + (y_0 - Cx_0)^\top V^{-1}(y_0 - Cx_0) + \sum_{k=1}^{h} \bar{w}_k^\top W^{-1} \bar{w}_k + (y_k - Cx_k)^\top V^{-1}(y_k - Cx_k)$$

subject to $x_k = A^{-1}x_{k+1} - A^{-1}\bar{w}_{k+1}$, for given positive definite $W, V, \bar{\Theta}_0$; then (ii) set $\bar{x}_h = \operatorname{argmin} J_h(x_h)$. (Interpretation: estimate $x_0, \dots, x_h$ by the values best "explained" by small disturbances, output noise, and deviation from the initial guess, then keep the last estimate.)

Show that $\bar{x}_h$ coincides with the Kalman filter estimate $\hat{x}_{h|h}$ obtained from the usual recursion (with $\Theta_{0|-1} = \bar{\Theta}_0$). [Suggestion: run the DP algorithm *forward* in time (a "cost-to-come" instead of a cost-to-go), and use the matrix inversion lemma $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ repeatedly.]

Solution

This is a duality result: the batch/DP estimation problem above is the "dual" of the standard LQR problem, running forward in time instead of backward. The proof shows, by induction on k, that if $J_k(x_k) = (x_k - \hat{x}_{k|k})^\top \Theta_{k|k}^{-1}(x_k - \hat{x}_{k|k}) + c_k$ for a constant c_k (true at $k = 0$ by construction, using the matrix inversion lemma to show $\Theta_{0|0}^{-1} = \bar{\Theta}_0^{-1} + C^\top V^{-1} C$), then minimizing over $\bar{w}_{k+1}$ in the recursive definition of J_{k+1} – itself a small quadratic minimization solved by completing the square – produces exactly the same form at $k+1$, with $\Theta_{k+1|k+1}$ and $\hat{x}_{k+1|k+1}$ obeying the *standard* Kalman filter update (prediction $\Theta_{k+1|k} = A\Theta_{k|k}A^\top + W$ followed by correction with gain $L_{k+1} = \Theta_{k+1|k}C^\top(C\Theta_{k+1|k}C^\top + V)^{-1}$). Carrying the induction to $k = h$ gives $J_h(x_h) = (x_h - \hat{x}_{h|h})^\top \Theta_{h|h}^{-1}(x_h - \hat{x}_{h|h}) + c_h$, whose minimizer is trivially $\bar{x}_h = \operatorname{argmin} J_h(x_h) = \hat{x}_{h|h}$, as desired. The full algebraic bookkeeping (several applications of the matrix inversion lemma to massage $A^{-\top}\Theta_{k|k}^{-1}A^{-1}$ terms into the standard $\Theta_{k+1|k}^{-1}$ form) is in SolutionsPS2.tex if you want to work through every line.

Problem 6.7 Consider the optimal control problem

$$\min \frac{1}{T+1}\mathbb{E}\Big[\sum_{k=0}^{T} x_k^\top \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} x_k + u_k^\top \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} u_k\Big]$$

for

$$x_{k+1} = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} u_k + w_k, \quad k \in \mathbb{N}_{\geq 0}$$

where $x_k, u_k \in \mathbb{R}^2$ and $w_k \in \mathbb{R}^2$ are iid random vectors.

(i) Find an optimal policy for u_0 and u_1 when $T = 2$.
(ii) Compute the expected optimal cost if $x_0 = [1 \;\; 2]^\top$ and $\mathbb{E}[w_k w_k^\top] = 3I$ for every k.
(iii) Find the optimal control policy for u_k, $k \in \mathbb{N}_{\geq 0}$, when $T \to \infty$.

Solution

Not reproduced here – taken from the TU/e 2015/2016 sample exam 2, question 3, whose full worked solution is available in the exam archive rather than in SolutionsPS2.tex. This is a direct application of the finite-horizon stochastic LQ result of Lecture 6 (slide "Optimal policy" – gains K_k from the same Riccati recursion as the deterministic case, with the expected cost picking up the extra $\operatorname{trace}(P_{k+1}\mathbb{E}[w_k w_k^\top])$ terms) for (i)-(ii), and of the averaged-cost infinite-horizon extension (Lecture 6, "Discussion" – cost converging to $\operatorname{trace}(\bar{P}W)$) for (iii); note $A = \operatorname{diag}(3/2, 1)$ is not itself stable, so (iii) is a genuine test of whether (A, B) is controllable enough for the algebraic Riccati equation to still apply.